Class 21: Fixed Income I

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Today's Question

UK pension funds nearly collapsed in 2022 because of... bonds? How does that happen?

Positioning

Where We've Been	Where We Are	Where We're Going
Alternatives (equity-focused)	Bond pricing, yields, and the term structure	Duration, Orange County, and interest-rate swaps

“UK Pensions Got Margin Calls”

Key themes: leveraged bond positions, liability-driven investing, and how “safe” assets can blow up

Learning Objectives

By the end of today's class, you should be able to:

1. Price zero-coupon and coupon bonds from a yield curve
2. Back out spot rates from coupon-bond prices (YTM vs. spot)
3. Derive forward rates by no-arbitrage
4. Interpret the slope of the yield curve under expectations and liquidity-preference theories

Zero Coupon Bonds

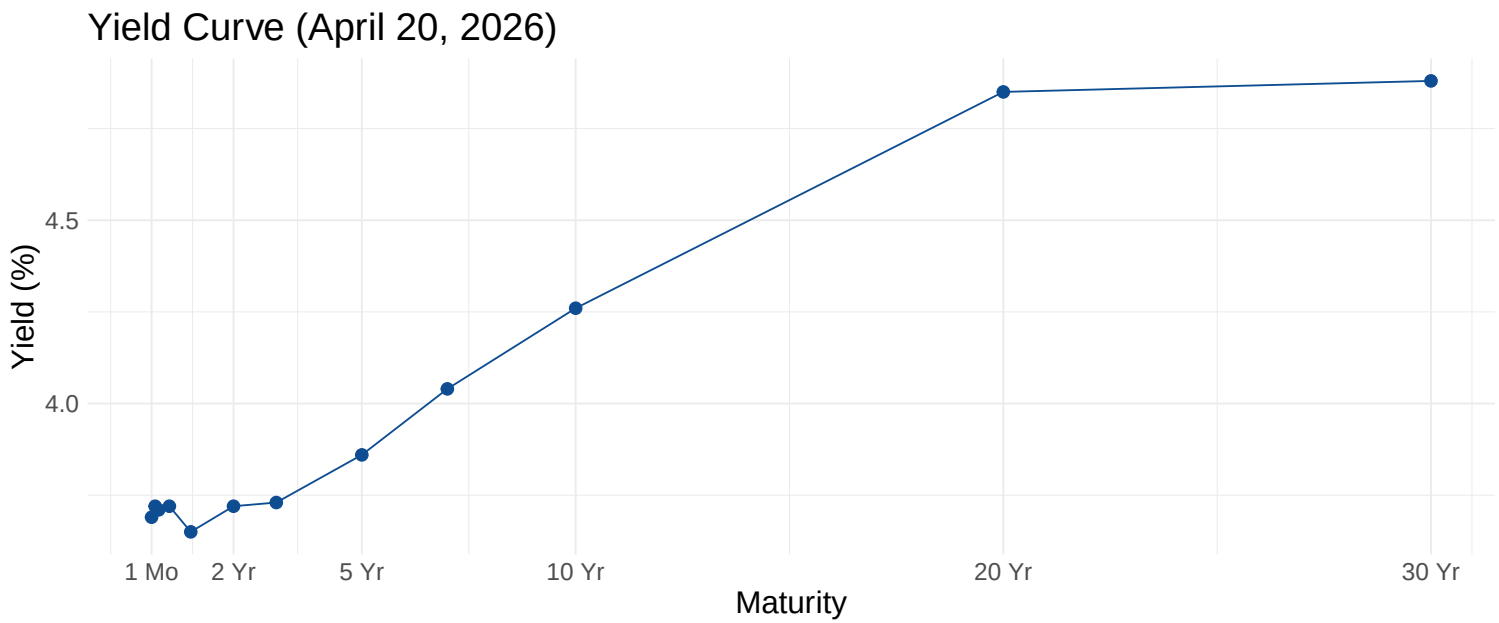
Consider a *risk-free* **zero-coupon** bond

- Face value (final payoff) of F with a maturity of T years
- If the interest rate for maturity T is r_T, then the current price of this bond is:

$$P_0 = \frac{F}{(1 + r_T)^T}$$

Yield Curve

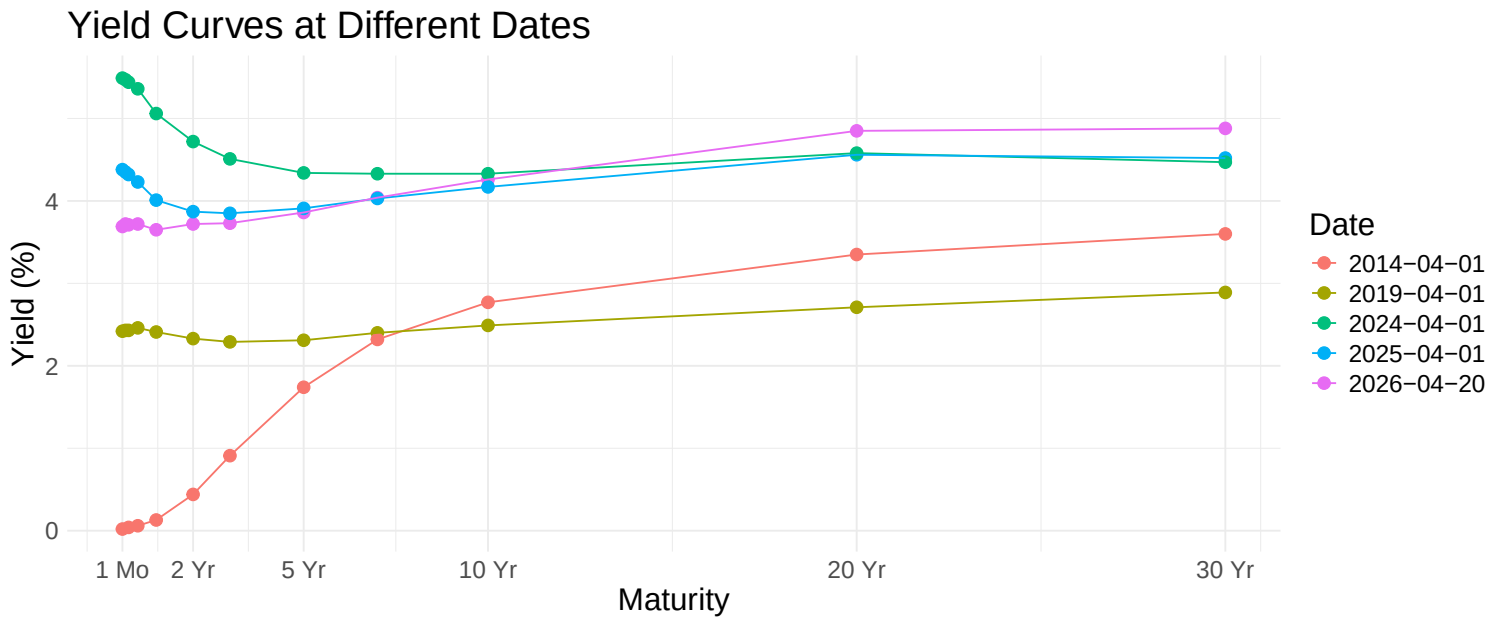
- Notation r_T implies variation in r over time
- **Yield curve** plots the term structure:



- In theory, produced from zero-coupon bonds or stripped Treasuries

Yield Curves Over Time

- The yield curve changes shape over time:



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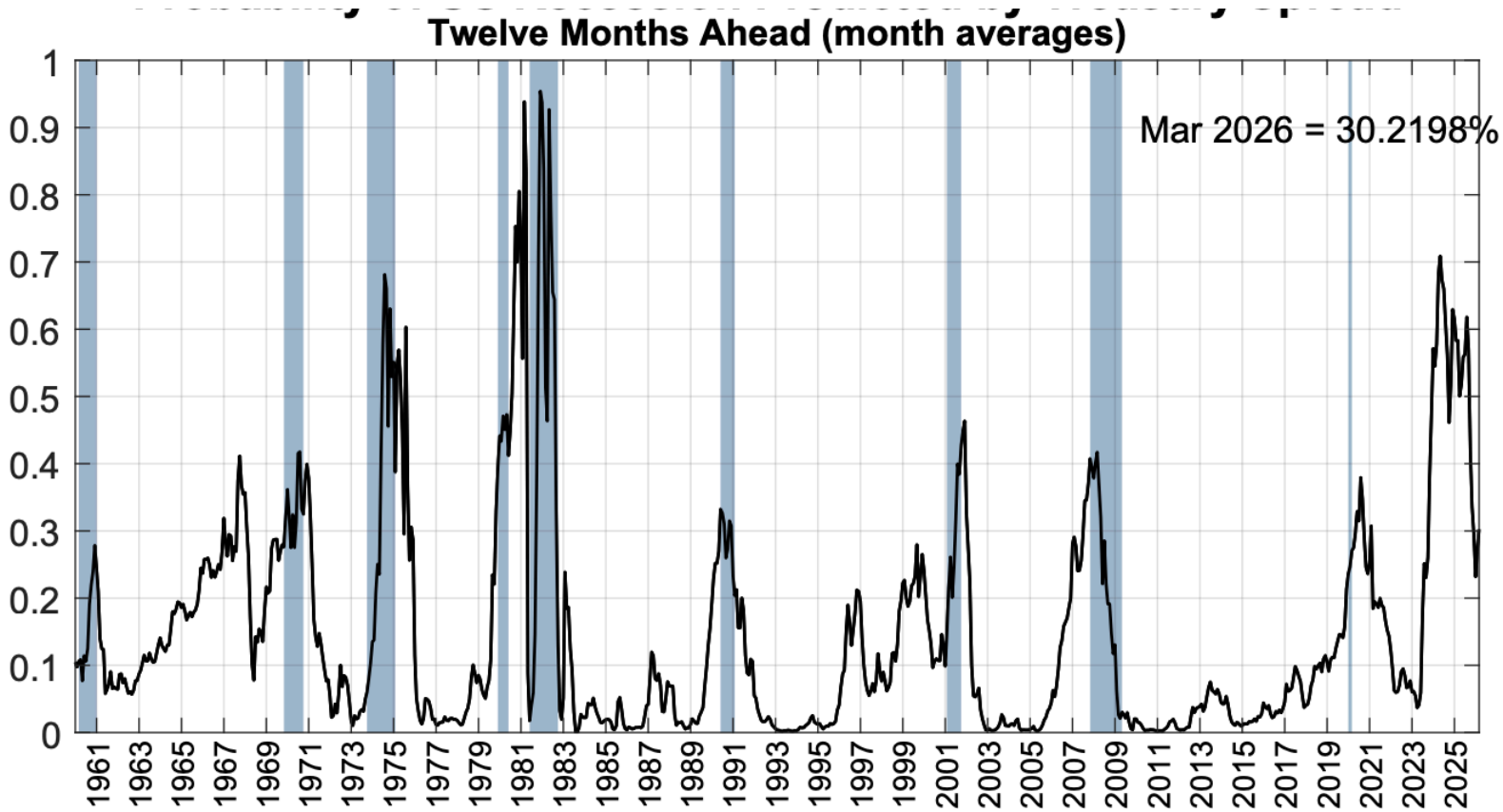
Yield Curve



*Parameters estimated using data from January 1959 to December 2009, recession probabilities predicted using data through Mar 2025.
The parameter estimates are α=-0.5333, β=-0.6330.

Updated 04-Apr-2025

Coupon Bonds

Coupon bonds pay coupon C in intervening periods (assume annually):

$$P_0 = \sum_{t=1}^{T} \frac{C}{(1+r_t)^t} + \frac{F}{(1+r_T)^T}$$

- C: interest or coupon payments
- T: number of periods (years) to maturity
- r_t: prevailing discount rate for horizon t

Yield to Maturity

- Suppose we observe coupon C, face value F and price P of a single bond
- Yield-to-maturity (YTM) solves for the discount rate that makes this price "right":

$$P = \sum_{t=1}^{T} \frac{C}{(1+\mathrm{YTM})^t} + \frac{F}{(1+\mathrm{YTM})^T}$$

- YTM is a weighted average of r_1 through r_T
- Plots of YTM by horizon approximate the true zero coupon yield curve

Yield to Maturity — Example

Consider a 10-year bond with face value of 100 and coupon payments of \$5/year

If these bonds trade **below** par, YTM exceeds coupon rate:

$$92.64 = \sum_{t=1}^{10} \frac{5}{(1+0.06)^t} + \frac{100}{(1+0.06)^{10}}$$

If bond trades at a **premium**, coupon rate exceeds discount rate:

$$108.11 = \sum_{t=1}^{10} \frac{5}{(1+0.04)^t} + \frac{100}{(1+0.04)^{10}}$$

Coupon Bonds and Yield Curves

- YTM collapses r_1 through r_T to a singular discount rate
- Suppose we want to solve for the entire yield curve
- If we want to know T interest rates over the yield curve, we need T bonds (roughly)
  - T equations and T unknowns
- Once we have interest rates for entire yield curve, can price any risk-free bond

Coupon Bonds and Yield Curves — Example

Use three bonds to solve for r_1, r_2, and r_3:

$$P_1 = \frac{C_1 + 100}{1 + r_1}$$

$$P_2 = \frac{C_2}{1 + r_1} + \frac{C_2 + 100}{(1 + r_2)^2}$$

$$P_3 = \frac{C_3}{1 + r_1} + \frac{C_3}{(1 + r_2)^2} + \frac{C_3 + 100}{(1 + r_3)^3}$$

Forward Rates

- Consider a contract in which we agree to lend money for 1 year, beginning 1 year in the future (repayment 2 years in the future)
  - $f_{1,2}$ is the *forward* rate for a 1-year loan issued 1 year from today
- Generally, $f_{j,k}$ is the forward rate from period j to k
  - $f_{0,1}$ is the normal 1-year spot rate, r_1
- A comment on notation: we use r_3 to mean today's prevailing interest rate for a 3-year horizon. This is different from BKM (in which r_3 refers to the 1-year rate which will prevail three years from now)

Forward Rates

Now consider two *equivalent* strategies:

1. Buy \$1 worth of a two-year zero coupon bond
   - Receive $(1 + r_2)^2$
2. Buy \$1 worth of a one-year zero coupon bond and a forward contract for 1 year beginning 1 year in the future
   - Receive $(1 + r_1)(1 + f_{1,2})$

By no arbitrage, since the bonds and contracts are riskless:

$$(1 + r_1)(1 + f_{1,2}) = (1 + r_2)^2$$

This implies that the 1- and 2-year bond rates set the forward rate:

$$f_{1,2} = \frac{(1 + r_2)^2}{1 + r_1} - 1$$

Forward Rates — General Case

- Extending to the general forward rate from t to T:

$$(1+r_T)^T = (1+r_t)^t (1+f_{t,T})^{T-t}$$

$$f_{t,T} = \left[\frac{(1+r_T)^T}{(1+r_t)^t}\right]^{\frac{1}{T-t}} - 1$$

For example, $r_1 = 5\%$, $r_2 = 6\%$:

$$f_{1,2} = \frac{(1.06)^2}{1.05} - 1 = 7.01\%$$

(When $T - t = 1$, this reduces to the formula from the previous slide.)

Forward Rates and Term Structure — Risk-Neutral

- Suppose investors are risk-neutral (or the future is certain)
- Then, forward rates should equal expectations of future short rates (why? — what trade could you do if not?)
  - $f_{1,2} = E(r_{1,2})$
  - $f_{2,3} = E(r_{2,3})$
- Hence, long-term rates are a geometric average of current and expected future short rates

$$(1 + r_2)^2 = (1 + r_{0,1})(1 + f_{1,2}) = (1 + r_{0,1})E(1 + r_{1,2})$$

Notation: $r_{t,t+1}$ denotes the (random) one-year spot rate that will prevail at time t; $E(r_{1,2})$ is today's expectation of that future rate.

Term Structure — Risk-Averse 2-Period Investor

- Now what if the investor is **risk-averse** and investing over two periods
- Faces a choice between a *sure* two-period rate, or a roll-over of a short-term bond
- Recall that a risk-averse investor prefers a sure thing if expected returns are equal
- In equilibrium, two-period investors bid **down** the long-term rate:

$$(1+r_2)^2 = (1+r_{0,1})(1+f_{1,2}) < (1+r_{0,1})E(1+r_{1,2})$$

Term Structure — Risk-Averse 1-Period Investor

- More commonly, consider a **risk-averse** *one-period* investor (needs liquidity at $t = 1$)
- Faces a similar choice: a *sure* one-period rate, or a two-period bond sold at $t = 1$

  1. Buy 1-year bond: $(1 + r_1)$
  2. Buy 2-year bond and sell at $t = 1$ at price $\frac{(1+r_2)^2}{1+E(r_{1,2})}$ — risky

- In equilibrium, one-period investors bid **down** the short-term rate:

$$(1 + r_2)^2 = (1 + r_{0,1})(1 + f_{1,2}) > (1 + r_{0,1})E(1 + r_{1,2})$$

Which Effect Dominates?

Two models, opposite predictions. Who wins empirically?

- Most bondholders — banks, money-market funds, corporates managing cash — care about **near-term liquidity**, not 30-year horizons
- So the **one-period investor** tends to dominate
- This generates an upward-sloping yield curve on average, with forward rates biased **above** expected future short rates
- Gap between $f_{t,t+1}$ and $E\left(r_{t,t+1}\right)$ is the **liquidity premium**

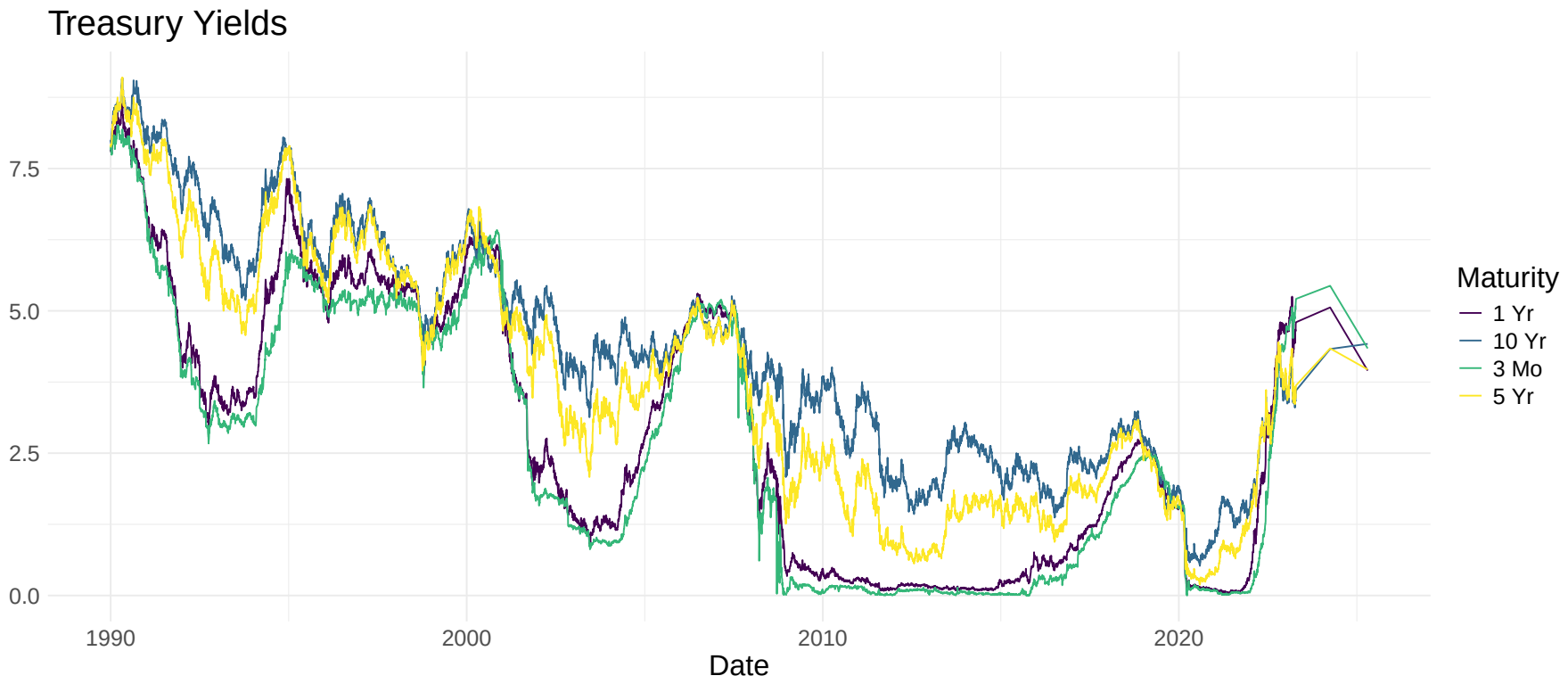
Interpreting Yield Curves

- Upward sloping yield curves are commonly interpreted as evidence of:
  1. Higher future expected short rates
  2. Liquidity preference for short-term bonds
- Define the liquidity premium as the gap between the forward rate and the expected future short rate:

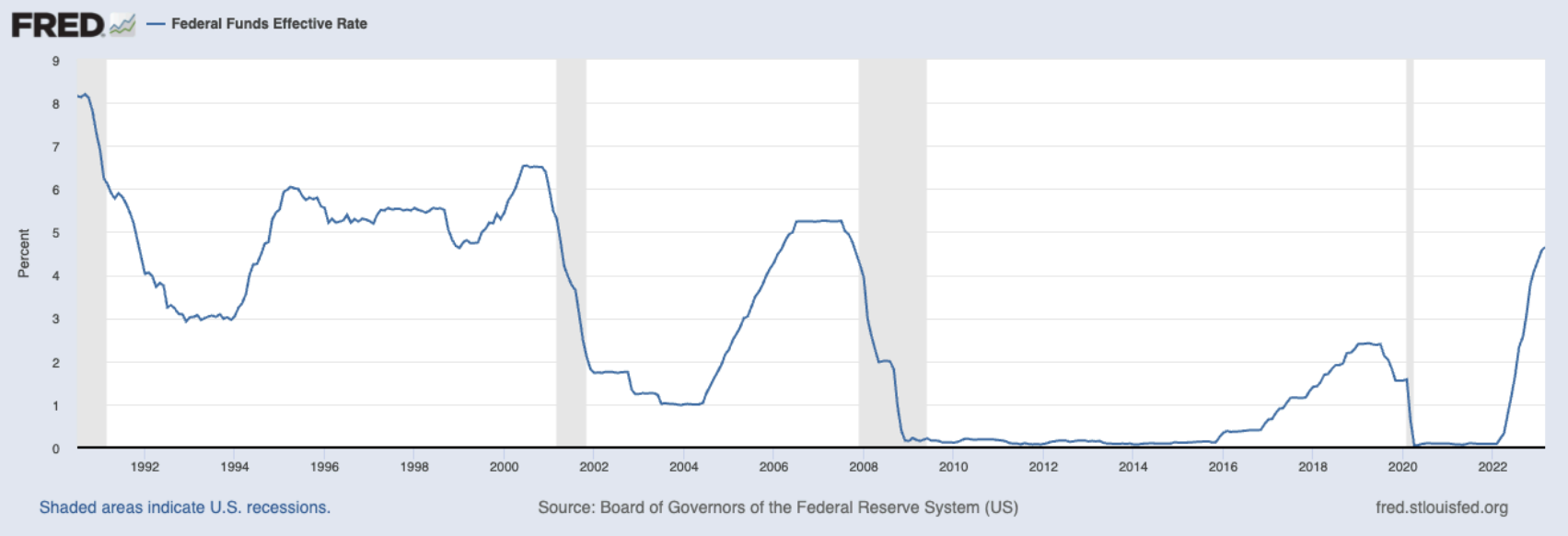
$$K = (1 + f_{t,t+1}) - E(1 + r_{t,t+1})$$

- $K > 0$ is exactly the one-period-investor story: forwards **overstate** expected short rates because short-horizon investors demand a premium to hold longer bonds
- Different combinations of expectations and K generate the yield-curve shapes we observe

Treasury Yields Over Time



FRED: Treasury Yield Curve



Next Class: April 21

- Topics: Fixed Income II
  - Bond pricing relationships
  - Duration and modified duration
  - Duration matching and immunization
  - Orange County (1994) case
  - Interest-rate swaps

- Matt Levine reading: “Startup Bank Had a Startup Bank Run”